

State of Developer Ecosystem 2024: Language Trends, Satisfaction, and Strategic Shifts

Key insights from JetBrains' survey of 23,262 developers worldwide on programming language usage, satisfaction, adoption intent, and domain specialization.

23,262

developers surveyed
Global community snapshot

2017–2024

usage trend window
Eight-year strategic view

2025

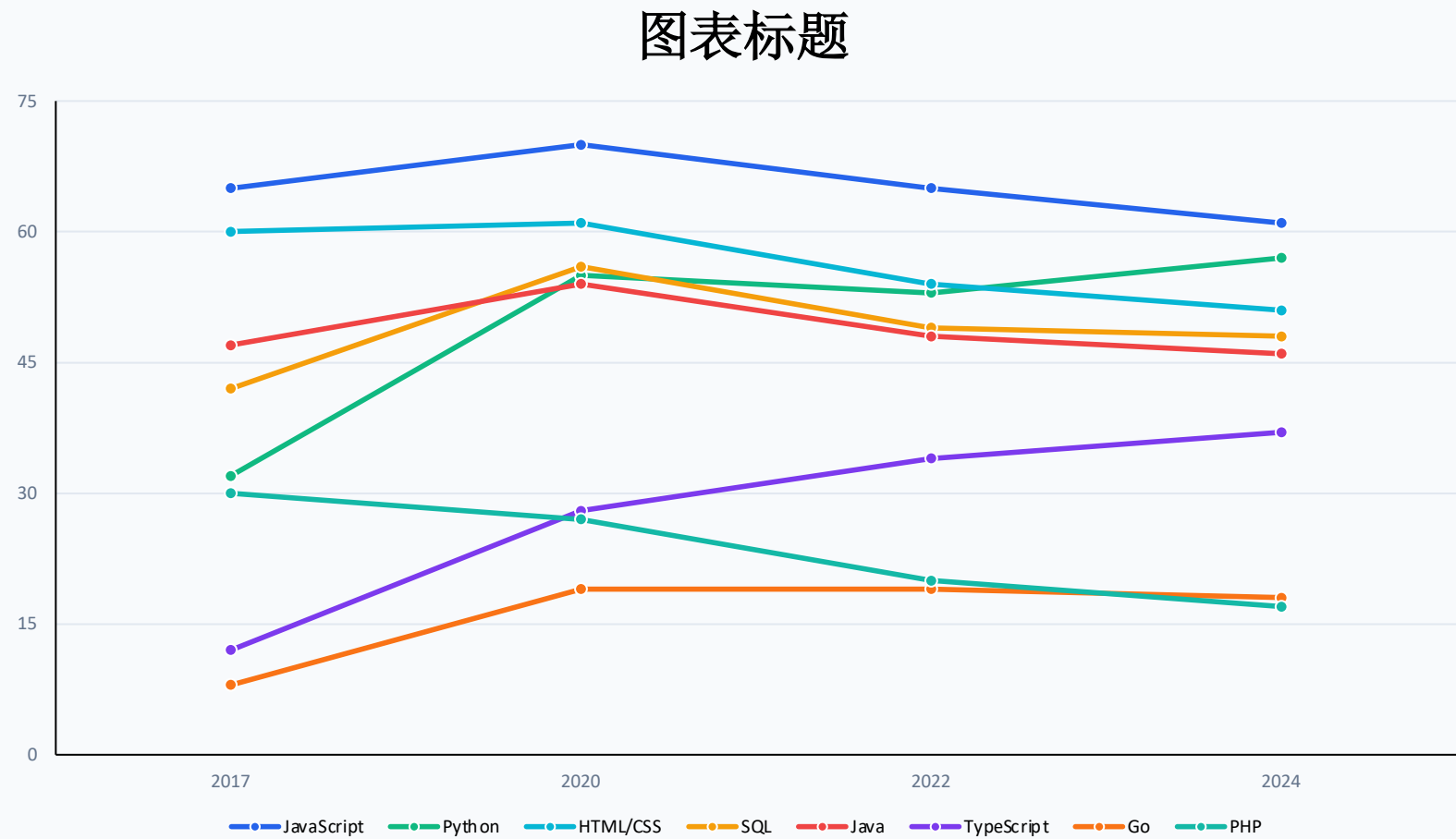
strategy lens
Hiring and platform priorities

Narrative arc

1. How usage is changing
2. What developers prefer
3. What to do about it

Python Closes the Gap: 8 Years of Language Usage Shifts

Usage share by language from JetBrains survey snapshots; all values are percentages of developers using each language.



JavaScript remains #1 by usage

61% in 2024, down from a 70% peak in 2020.

Python nearly closes the gap

32% in 2017 → 57% in 2024 as AI/data work expands.

TypeScript triples

12% in 2017 → 37% in 2024; frontend stacks are typed.

PHP declines materially

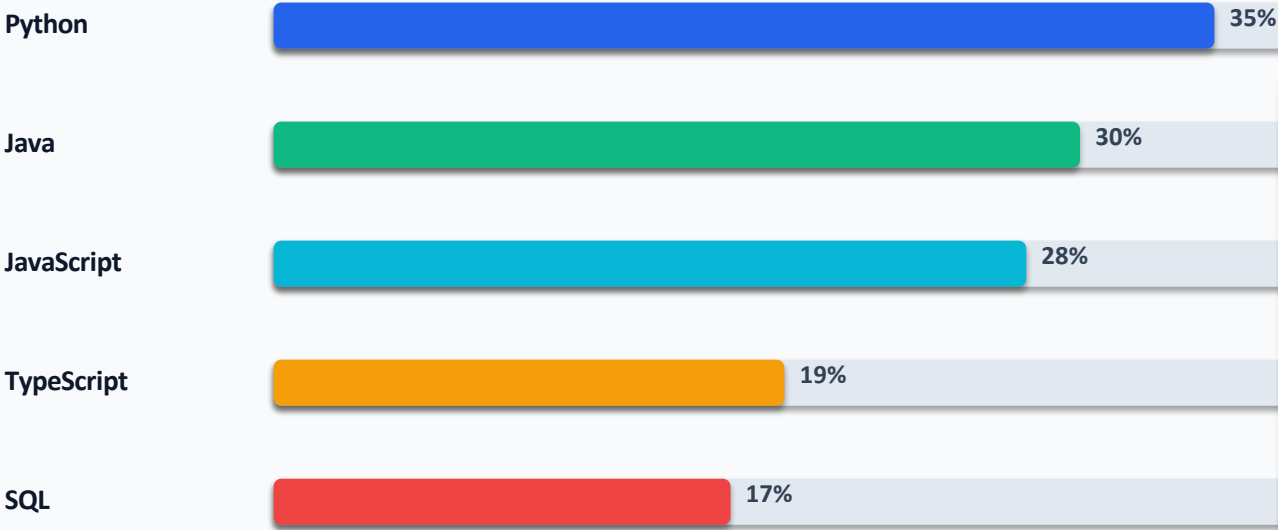
30% in 2017 → 17% in 2024, signaling a shrinking pool.

2024 TOP 5

JavaScript 61% | Python 57% | HTML/CSS 51% | SQL 48% | Java 48%

Python Is Now the #1 Primary Language at 35%

Primary-language concentration is high, while known adoption intent remains low across the listed languages.



Primary-language share

Comfort-zone effect

Developers rarely plan to switch primary stacks.



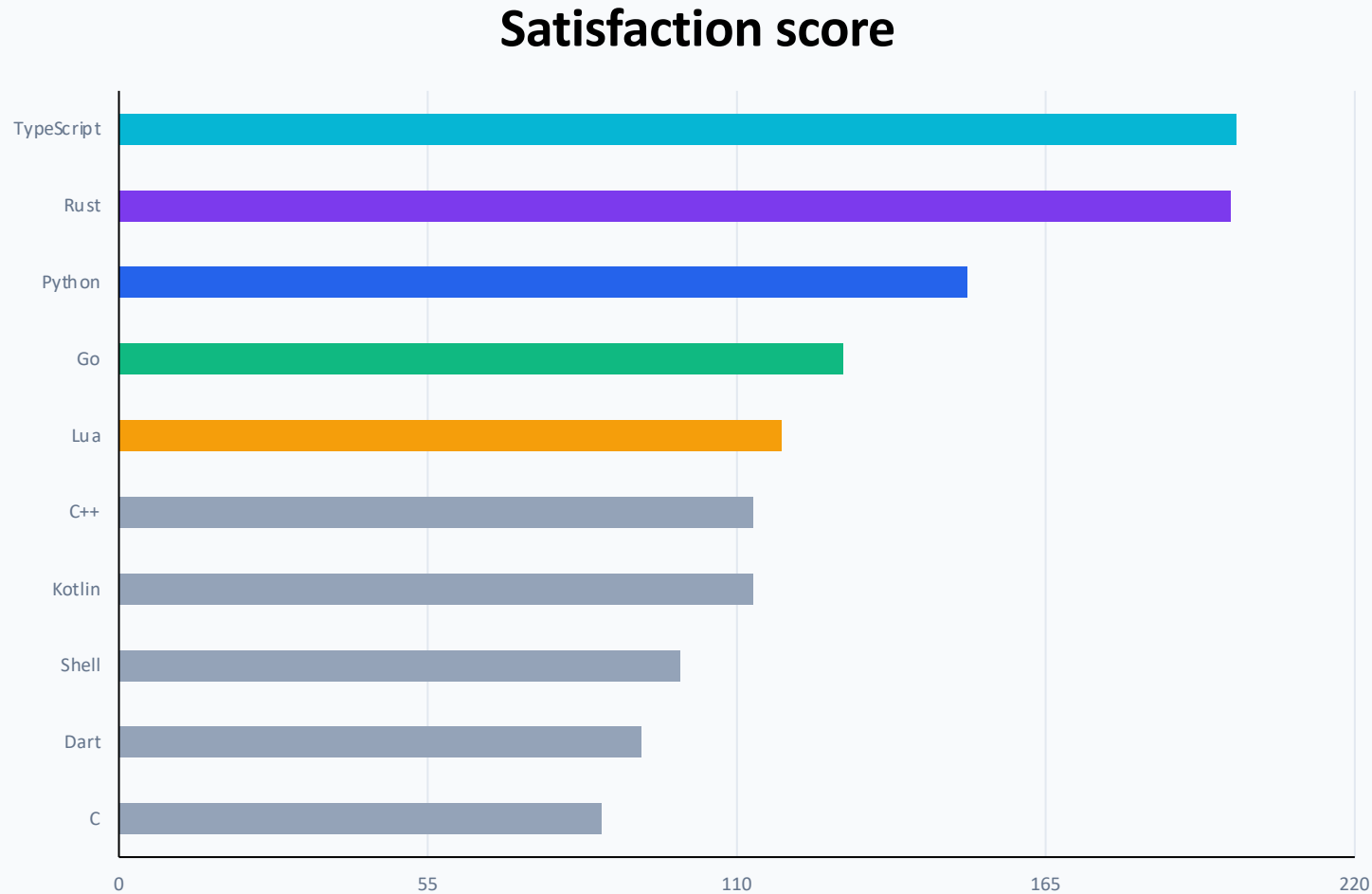
Known adoption intent in the brief is 6% or lower, reinforcing ecosystem lock-in and expertise investment.

Strategic implication

Hiring and enablement plans should optimize around existing language ecosystems, not assume broad migration intent.

TypeScript and Rust Top Developer Satisfaction, Nearly Tied at 199 vs 198

Satisfaction score ranking for the top 10 languages named in the JetBrains brief.



Type safety wins mindshare

TypeScript (199) and Rust (198) sit virtually tied at the top.

Python remains highly loved

A score of 151 pairs broad usage with strong satisfaction.

Tooling affects retention

Modern tooling, safety, and expressive type systems correlate with enthusiasm.

Each Language Owns a Domain: AI Belongs to Python, UI to TypeScript

Leading language by application domain; percentages are domain selections reported in the brief.

Domain	Lead language	Share	Strategic read
AI / Machine Learning	Python	31%	Specialize AI hiring and tooling around Python.
Data Processing & Analytics	Python	17%	Keep Python central in analytics platforms.
User Interface	TypeScript	62%	Standardize typed UI stacks with TypeScript.
APIs & Services	Java	49%	Java remains a strong service-platform default.
Application Logic & Workflows	Java	55%	Enterprise workflow logic still favors Java.
System Tools & Components	C++	27%	C++ leads low-level systems components.
Integrating with APIs & Services	Python	38%	Python is a flexible integration layer.

UI is a two-language stronghold

TypeScript leads UI at 62%; JavaScript is also dominant at 58%.

Strategy principle

Language choice is domain-driven; optimize by workload, not global preference.

What This Means for Your Technology Strategy in 2025

Prioritize platform investment, hiring, and retention around domain fit and developer satisfaction.

Invest in Python for AI/ML teams

Python leads AI/ML at 31%, data science at 28%, and primary language share at 35%.

Make TypeScript the default UI bet

TypeScript leads UI at 62% and tops satisfaction at 199.

Use Rust and Go as infrastructure growth plays

Rust continues climbing to 11%; Go stabilized around 18% after rapid growth.

Plan deliberately for shrinking legacy pools

PHP fell from 30% in 2017 to 17% in 2024; Ruby fell from 10% to 4%.

Treat developer experience as retention strategy

Type-safe languages with modern tooling dominate satisfaction rankings.

Decision rule: align languages to domain ownership, then fund tooling and hiring where satisfaction and strategic demand overlap.